Lesson 1.5 — Domain and Range

Big idea: ***Domain*** *= every input the function is allowed to eat.* ***Range*** *= every output it can spit back out. Most domain problems boil down to three forbidden moves.*

Quick review

Three things you can't do: divide by zero, take the square (or any even) root of a negative, take the log of zero or a negative number.

From an equation: set up inequalities so each forbidden move is avoided, then solve.

From a graph: domain = the x-values the graph covers; range = the y-values it covers. Watch open vs. closed dots and arrows.

Notation: interval form (e.g., $[3, \infty)$) is fastest; set-builder form ($\{x \mid x \geq 3\}$) reads cleanly in proofs.

Worked example

Worked example. Find the domain and range of $f(x) = \sqrt{(x - 3)} + 2$.

Domain: $x - 3 \geq 0 \Rightarrow x \geq 3$, i.e. $[3, \infty)$.
Range: $\sqrt{(\ldots)}$ gives outputs ≥ 0, then we add 2, so $y \geq 2$, i.e. $[2, \infty)$.

Warm-ups

1. Find the domain of $f(x) = 2x^2 - 5x + 1$.
2. Find the domain of $g(x) = 1 / (x - 4)$.
3. Find the domain of $h(x) = \sqrt{(x + 7)}$.

Practice

1. Find the domain of $f(x) = (x + 1) / (x^2 - 9)$.

2. Find the domain and range of $g(x) = (x - 2)^2 - 4$.

3. From the graph description (a parabola opening up with vertex at $(1, -5)$, continuous everywhere), state the domain and range.

Challenge

1. Find the domain of $f(x) = \sqrt{(9 - x^2)} / (x - 1)$.

2. A box has square base of side x and height $10 - x$. Write the volume $V(x)$, then state the realistic domain (no negative or zero dimensions) and explain why the algebraic domain is wider than the real-world one.

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Answer key — Lesson 1.5

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. All real numbers, $(-\infty, \infty)$.
2. All real numbers except 4: $(-\infty, 4) \cup (4, \infty)$.
3. $x + 7 \geq 0 \Rightarrow x \geq -7$, i.e. $[-7, \infty)$.

Practice

1. $x^2 - 9 \neq 0 \Rightarrow x \neq \pm 3$. Domain: $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$.
2. Domain: all reals. Range: $y \geq -4$, i.e. $[-4, \infty)$.
3. Domain: all reals. Range: $[-5, \infty)$.

Challenge

1. Need $9 - x^2 \geq 0 \Rightarrow -3 \leq x \leq 3$, AND $x \neq 1$. Domain: $[-3, 1) \cup (1, 3]$.
2. $V(x) = x^2(10 - x)$. Algebraic domain: all reals. Realistic domain: $0 < x < 10$ (need positive side and positive height). The algebra doesn't know boxes can't have negative sides; you have to add the constraint from context.

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